

Dynamics from Probability: Predictive Operators and the Koopman Picture

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Abstract

Paper I produced a probability measure P on the observable σ -algebra from coherent structured observations. This paper asks what temporal structure P carries: once the observer's predictions are fixed, what are the laws of motion?

The answer is extracted entirely from P without additional assumptions. Given two measurable maps $Q : \Omega \rightarrow \alpha$ (present) and $F : \Omega \rightarrow \beta$ (future), the *predictive kernel* Π_Q is the regular conditional distribution of F given Q . The *minimal predictive state map* $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$ packages this conditional law as a single measurable map; the predictive factorization theorem shows that every conditional expectation of a function of F factors through Q_* . Indexing over a discrete time horizon, the kernels $\{\Pi_t\}$ compose via the Chapman–Kolmogorov equation, which is derived from temporal coherence of prediction rather than imposed. This produces the semigroup law $K_{t+s} = K_t \circ K_s$ for the *predictive operators* $\{K_t\}$, and the Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ connects operator-on-functions with measure-on-states. When the kernels are Dirac measures, $K_t g = g \circ \varphi_t$ and the classical Koopman operator is recovered. The whole picture — predictive state space, semigroup, Koopman–Perron duality, deterministic limit — assembles into a single structure called the *observable dynamical system*, which Paper III will use as the object to be reconstructed from time-delayed observations. The main theorems are formalized in Lean 4 / Mathlib with no outstanding proof obligations.

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1 Introduction

Paper I produced a probability measure P on the observable σ -algebra from coherent structured observations; collective exhaustion was the exact condition licensing the passage from finite coherence to genuine probability. The question this paper addresses is what temporal structure P carries. Once the probability is fixed, the laws of motion are already determined.

The classical approach to dynamics, following Koopman [1], describes how observable functions evolve: the operator $U_T f = f \circ T$ acts by precomposing with the dynamics T . This presupposes access to T . The approach here replaces that assumption with a prior question: given the current observation $Q : \Omega \rightarrow \alpha$, what does the observer predict about the future observation $F : \Omega \rightarrow \beta$? The answer is the *predictive kernel* Π_Q , the regular conditional distribution of F given Q . No state-space transformation is assumed.

The predictive kernel packages into the *minimal predictive state map* $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$, sending each sample point to its conditional law. The predictive factorization theorem shows that every conditional expectation of a function of F factors through Q_* : it is the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition law is the Chapman–Kolmogorov equation, derived from temporal coherence, not assumed. This yields a semigroup of *predictive operators* $\{K_t\}$ and the Koopman–Perron duality

$\int K_t g d\mu = \int g d(P_t^* \mu)$. When the kernels are Dirac measures, $K_t g = g \circ \varphi_t$ recovers the classical Koopman operator.

These structures assemble into the *observable dynamical system* $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$, which Paper III takes as its starting point and asks: when does the delay orbit of a single function reconstruct the full state space?

2 Setup

The paper is self-contained. The connection to Paper I [2] is motivational; the formalism starts here from a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [3] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The *joint pushforward* is the measure $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$; its marginals are $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The main theorems are formalized in Lean 4 / Mathlib [4] in `PredictiveState.lean` and `PredictiveOperators.lean`. Several further results — the experiment preorder, general predictive sufficiency, the continuous-time generator — are stated in the paper but not yet formalized; these are noted at the relevant points.

3 The predictive kernel and the minimal predictive state map

3.1 The predictive kernel

Definition 3.1 (Predictive kernel). The *predictive kernel* $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ is the regular conditional distribution of F given Q :

$$\Pi_Q(q, A) = P(F \in A \mid Q = q),$$

defined as the disintegration kernel $\Pi_Q(q, \cdot) = \rho(\cdot \mid \alpha)$ of the joint pushforward $\rho = P \circ (Q, F)^{-1}$ over its first marginal.

Concretely: Π_Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\alpha \rightarrow \mathcal{P}(\beta)$ such that

$$\int_A \Pi_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Proposition 3.2 (Predictive compatibility, P2.2). *Let $\pi : \alpha \rightarrow \alpha'$ be measurable with $Q' = \pi \circ Q$. Then for $(P \circ Q^{-1})$ -almost every $q \in \alpha$:*

$$\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot).$$

Proof. The joint pushforward $\rho' = P \circ (Q', F)^{-1}$ satisfies $\rho' = \rho \circ (\pi \times \text{id})^{-1}$. The claim follows from the $P \circ Q^{-1}$ -a.e. uniqueness of the disintegration of ρ' over its first marginal, combined with the covariance of `condKernel` under measurable maps (`eq_condKernel_of_measure_eq_compProd` in Mathlib). $\square$

3.2 The predictive law map and the minimal predictive state map

The predictive kernel Π_Q maps each value $q \in \alpha$ to a probability measure on outcomes. Packaging this as a single measurable map gives:

Definition 3.3 (Predictive law map and minimal predictive state map). The *predictive law map* is

$$\varphi_Q : \alpha \rightarrow \mathcal{P}(\beta), \quad \varphi_Q(q) = \Pi_Q(q, \cdot).$$

The *minimal predictive state map* is the composition

$$Q_* := \varphi_Q \circ Q : \Omega \rightarrow \mathcal{P}(\beta).$$

The value $Q_*(\omega)$ is the full conditional law $P(F \in \cdot \mid Q = Q(\omega))$.

Proposition 3.4 (Measurability, P2.1/P2.3). *The maps φ_Q and Q_* are measurable.*

Proof. φ_Q is the underlying map of the Markov kernel Π_Q , which is measurable by definition of a kernel (`Kernel.measurable` in Mathlib). Since $\mathcal{P}(\beta)$ carries the subspace σ -algebra from the Giry space of all measures on β , the wrap as a `ProbabilityMeasure` is measurable by `Measurable.subtype_mk`. Measurability of Q_* then follows from composition. $\square$

4 Predictive factorization

The predictive factorization theorem is the central result of this paper. It says that the conditional expectation of any bounded measurable function of F , given Q , factors through the minimal predictive state map Q_* .

Theorem 4.1 (Predictive factorization, P2.5). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \left(\omega \mapsto \int g dQ_*(\omega) \right) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega) = \int g d\Pi_Q(Q(\omega), \cdot)$. We verify the two defining properties of the conditional expectation.

Measurability. The map $q \mapsto \int g d\Pi_Q(q, \cdot)$ is strongly measurable (Bochner integrability of g against a Markov kernel; `StronglyMeasurable.integral_kernel` in Mathlib). Since $Q_* = \varphi_Q \circ Q$, the map $h = (\text{predictive operator applied to } g) \circ Q_*$ is $\sigma(Q)$ -measurable.

Set-integral identity. For every $\sigma(Q)$ -measurable set $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_{Q^{-1}(A)} \int g d\Pi_Q(Q(\omega), \cdot) dP(\omega) \\ &= \int_A \int g d\Pi_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{push-forward}) \\ &= \int_A \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration identity}) \\ &= \int_{Q^{-1}(A)} g(F(\omega)) dP(\omega) = \int_S g(F) dP. \end{aligned}$$

The key step is the disintegration identity for $\rho = P \circ (Q, F)^{-1}$, which in Mathlib is `Measure.setIntegral_cond`.

By the a.e. uniqueness of conditional expectation (`ae_eq_condExp_of_forall_setIntegral_eq`), $h = E[g(F) \mid \sigma(Q)]$ P -a.e. $\square$

Corollary 4.2 (Predictive sufficiency, P2.5). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e.

Proof. Since $Q_* = \varphi_Q \circ Q$, we have $\sigma(Q_*) \subseteq \sigma(Q)$. The conditional expectation given $\sigma(Q)$ already factors through Q_* by Theorem 4.1, so it is $\sigma(Q_*)$ -measurable. $\square$

The canonical predictive operator K_{Q_*} is the map

$$(K_{Q_*}g)(q_*) = \int g dq_*, \quad q_* \in \mathcal{P}(\beta).$$

Theorem 4.1 then reads: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e. This is the canonical operator associated with the predictive state.

5 The predictive semigroup

So far, the setup has been static: a single pair of measurable maps (Q, F) representing present and future. To introduce dynamics, we index over a discrete time horizon. For each $t \in \mathbb{N}$, let $F_t : \Omega \rightarrow \gamma$ be the observable at time t , and let Π_t be the predictive kernel of F_t given the current predictive state.

5.1 Predictive kernel families

Definition 5.1 (Predictive kernel family and semigroup kernel). A *predictive kernel family* on a state space γ is a family $\{\Pi_t\}_{t \in \mathbb{N}}$ of Markov kernels $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ with $\Pi_0 = \delta$ (identity: $\Pi_0(q, \cdot) = \delta_q$).

A *predictive semigroup kernel* is a predictive kernel family satisfying the Chapman–Kolmogorov equation:

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad \text{for all } q \in \gamma, t, s \in \mathbb{N}.$$

In kernel notation: $\Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s$ where $\circ_{\kappa}$ denotes kernel composition.

5.2 The predictive operator family

Definition 5.2 (Predictive operator family). Given a predictive kernel family $\{\Pi_t\}$, the *horizon-indexed predictive operator* at horizon t is

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Proposition 5.3 (Basic properties, P3.1/P3.4). *For any predictive kernel family $\{\Pi_t\}$:*

- (i) Identity at zero: $K_0 g = g$ for all measurable g ;
- (ii) Positive: $g \geq 0 \Rightarrow K_t g \geq 0$;
- (iii) Unital: $K_t \mathbf{1} = \mathbf{1}$;
- (iv) Contractive: $\|K_t g\|_{\infty} \leq \|g\|_{\infty}$.

Proof. Identity at zero: $K_0 g(q) = \int g d\delta_q = g(q)$. Positivity and unitality are immediate from the Markov property and non-negativity of integration against a positive measure. Contractiveness: $|(K_t g)(q)| = |\int g d\Pi_t(q, \cdot)| \leq \|g\|_{\infty} \cdot \Pi_t(q, \gamma) = \|g\|_{\infty}$. $\square$

Theorem 5.4 (Semigroup property, P3.3). *Let $\{\Pi_t\}$ be a predictive semigroup kernel. For every bounded measurable $g : \gamma \rightarrow \mathbb{R}$:*

$$K_{t+s} g = K_t (K_s g).$$

Proof. Using the Chapman–Kolmogorov equation $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$:

$$\begin{aligned}
(K_{t+s}g)(q) &= \int g \, d\Pi_{t+s}(q, \cdot) \\
&= \int g \, d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) \\
&= \int \left(\int g \, d\Pi_t(q', \cdot) \right) d\Pi_s(q, q') \quad (\text{Bochner–Fubini for kernel composition}) \\
&= \int (K_t g)(q') \, d\Pi_s(q, q') = (K_t(K_s g))(q).
\end{aligned}$$

The Bochner–Fubini step is `Kernel.integral_comp` from `Mathlib.Probability.Kernel.Composition.Integral`. The boundedness hypothesis on g is needed for integrability of g against the composed kernel. $\square$

6 Koopman–Perron duality

The predictive operators K_t act on functions. The dual objects act on measures.

Definition 6.1 (Koopman–Perron dual). Given a predictive semigroup kernel $\{\Pi_t\}$ and a (finite) measure μ on γ , the *Koopman–Perron push-forward* is

$$P_t^* \mu := \mu \star \Pi_t = \int_\gamma \Pi_t(q, \cdot) \, d\mu(q).$$

In Mathlib notation: $\mu.\text{bind } (\text{ker } t)$.

Theorem 6.2 (Koopman–Perron duality, P3.5). *For every bounded measurable $g : \gamma \rightarrow \mathbb{R}$ and finite measure μ on γ :*

$$\int_\gamma (K_t g) \, d\mu = \int_\gamma g \, d(P_t^* \mu).$$

Proof.

$$\begin{aligned}
\int g \, d(P_t^* \mu) &= \int g \, d(\mu \star \Pi_t) = \int g \, d\left(\int \Pi_t(q, \cdot) \, d\mu(q)\right) \\
&= \int_\gamma \int g \, d\Pi_t(q, \cdot) \, d\mu(q) \quad (\text{Bochner–Fubini}) \\
&= \int_\gamma (K_t g)(q) \, d\mu(q).
\end{aligned}$$

The Bochner–Fubini step again uses `Kernel.integral_comp`. $\square$

7 Deterministic specialization

When the dynamics is deterministic, the stochastic picture collapses to the classical Koopman picture.

Theorem 7.1 (Deterministic specialization, P3.6). *Suppose $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$. Then for every measurable g :*

$$K_t g = g \circ \varphi_t.$$

Proof. $(K_t g)(q) = \int g \, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The integral against a Dirac measure is `integral_dirac` in Mathlib. $\square$

Corollary 7.2 (Flow law, P3.6). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for all t , then*

$$\varphi_{t+s} = \varphi_t \circ \varphi_s.$$

Proof. From $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ and the Dirac hypothesis:

$$\begin{aligned} \delta_{\varphi_{t+s}(q)} &= \Pi_{t+s}(q, \cdot) = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) \\ &= \Pi_s(q, \cdot) \star \Pi_t = \delta_{\varphi_s(q)} \star \Pi_t = \Pi_t(\varphi_s(q), \cdot) = \delta_{\varphi_t(\varphi_s(q))}. \end{aligned}$$

Since Dirac measures at distinct points are distinct (under **SeparatesPoints**), we conclude $\varphi_{t+s}(q) = \varphi_t(\varphi_s(q))$. $\square$

8 The observable dynamical system

The results above assemble into a single structure.

Definition 8.1 (Observable dynamical system). An *observable dynamical system* is a triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where:

- $O_{Q_*} = \mathcal{P}(\beta)$ is the *predictive state space* (the image of Q_*);
- $\nu_{Q_*} = P \circ Q_*^{-1}$ is the *predictive state distribution* (the pushforward of P along Q_*);
- $\{K_t\}$ is the *predictive operator family* on O_{Q_*} , satisfying the semigroup law and Markov properties above.

9 Relation to Paper III

Paper II has constructed the dynamics from the probability measure P and the measurable maps (Q, F) . The output is the observable dynamical system $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$: a predictive state space, a distribution, and a semigroup. This raises the inverse question.

The construction specialises directly to the setting of Paper III. If $(X, \mathcal{B}, \mu, T)$ is a measure-preserving system and $h \in L^\infty$, set $Q_t = h \circ T^t$. Then $Q_* = \Phi_h$, the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the predictive state space O_{Q_*} is precisely the delay orbit space of Paper III.

Paper III asks: *when does the sequence of time-delayed observations $\{h \circ T^n : n \in \mathbb{Z}\}$ of a single function h determine the full state space?* The answer is the *reconstruction theorem*: for a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and $h \in L^\infty$, the observable σ -algebra $\mathcal{O}_h = \sigma(\{h \circ T^n\})$ equals $\mathcal{B}(X)$ modulo μ if and only if the algebra generated by $\{h \circ T^n\}$ is dense in $L^2(X, \mu)$. The proof turns on the *density bridge*: the L^2 -closure of a function algebra equals the L^2 -space of the σ -algebra it generates.

The Koopman operators $\{U_T^t\}$ on $L^2(X, \mu)$ connect the two papers. Paper II produces $\{K_t\}$ on the predictive state space O_{Q_*} ; when Q_* is injective modulo P -null sets, K_t and U_T^t act on the same information. The *cyclic vector condition* — that $\overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu)$ — appears naturally in the Koopman picture and is a sufficient condition for reconstruction. But it is strictly stronger: reconstruction requires only that the *algebra* generated by the delay orbit is L^2 -dense, whereas cyclicity requires the *linear span*. They coincide only when U_T has simple L^2 -spectrum.

When reconstruction holds, the predictive state space O_{Q_*} is, up to measure-theoretic identification, the full state space X . The compact Stone space $\text{St}(\mathcal{C})$ built in Paper I as a technical device for measure extension is identified by Paper III, under the reconstruction condition, with X itself. The programme begins with a compact completion of the sample space and ends by recognising it as the state space.

Remark 9.1 (Continuous-time deferred). All results above are stated for discrete time $t \in \mathbb{N}$. The continuous-time ($t \in \mathbb{R}_{\geq 0}$) versions require the infinitesimal generator of a C_0 -semigroup and the Hille–Yosida theorem. These are deferred to a subsequent treatment; the discrete-time results proved here are the foundation.

References

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